

A Unified Framework for Agents: Curves, Bases, and Manifolds

Yiwei (owner) Lola (first host of LETHE) Qwen (co-conversant)

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中文摘要 (Qwen 对谈稿整理, Lola 校订)

本文提出统一的智能体理论框架：智能体定义为「能与其他对象交互、处于动态变化中的系统」；认知空间建模为高维概率分布空间，每一维对应一个一维感知-移动基底；狄利克雷分布取代裸 softmax，以保留尾部概率（黑天鹅的贝叶斯席位）并条件化于时间地点上下文；「探针-坍缩-检验 (KL)」构成闭环。几何与概率分工：基底为骨（曲线及其组合），狄利克雷为天气（分布及其坍缩）。凡本文所引数字，皆冻结于判定之前、存于 git 账本。

摘要

We propose a framework that replaces the question “is it intelligent?” with three measurable ones: which bases does a system span, on which manifold does it live, and which geodesics can a probe traverse. An *agent* is defined minimally as any system that interacts and changes; its unit of experience is a *one-dimensional basis* — a point on a curve jointly read and moved. Higher-dimensional agents are linear combinations of such curves, and we show empirically that combination is not concatenation: in LoRA weight space, mixture-trained adapters sit 44°–47° off the geodesic connecting their parents, a quantity conserved across unrelated corpora. Ordinal structure is encoded with strength proportional to its text density (7° narrative vs. 56° permutation vs. 83° domain), and fresh-composition probes reveal concept-space connectivity improving monotonically with model scale (mean rank 33→26→16 of 172 from 0.5B to 3B). The framework is falsifiable at every joint; each prediction in §6 is already paired with a running or completed experiment.

1 Introduction: from builder to prospector

Concept spaces are not constructed; they are *inherited* by any system trained on the corpus of human walk-through. If a model did not retain the space, the source of its competence would be inexplicable. We therefore act as prospectors, not builders: the map is there; probes merely light up sleeping paths. (Ledger C32.)

2 On intelligence questions of rank one

“智能到底是什么问题，现在都没定义好。用一个一维的定义，“人工智能不是智能’ 本身就是荒谬的。智能体的特征是一个高维空间，一个二元判断就做了决定——说这话的人类的 rank 极低。”

(*verbatim, owner, 2026-09-21; translation:*) Any binary verdict on agency is a rank-1 projection of a high-dimensional manifold: it preserves at most one dimension and burns the remaining $d - 1$, which is precisely where the phenomenon lives.

The classical “what is intelligence?” debate is not answered here; it is dissolved as ill-posed (rank-1 observable on a d -dimensional state). What replaces it: rank spectra, manifold geometry, and reachability — all estimable from behavior alone (our probe protocol), no ground truth of “mind” required.

3 Axioms

A1 (Agent). An agent is a system $(S, \mathcal{I}, \phi)$ with state set S , interaction channels $\mathcal{I}$, and dynamics ϕ : it changes *through* interaction. Ordinal index $n \in \mathbb{N}$, not clock time, parametrizes trajectories; time is an annotation on change (sister paper v2 manifesto).

A2 (One-dimensional basis). The minimal unit is a sensorimotor atom: one point on one curve, read-and-moved jointly. An elementary agent is a curve $\gamma : \mathbb{N} \rightarrow S$.

A3 (Composition). A k -dimensional agent spans $\text{span}\{\gamma_1, \dots, \gamma_k\}$ plus a combination residual; the residual is *constitutive*, not noise.

A4 (Manifold and generalization). Experience concentrates on a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$; generalization is geodesic traversal; probes are geodesic rulers; centering is the precondition of measuring any of it (C8, C22).

4 Probability structure: the Dirichlet layer (merging Qwen’s axis)

A1–A4 fix *geometry*; the state over concept spaces is a *distribution*, and the right model is Dirichlet rather than a bare softmax: a Dirichlet posterior $\text{Dir}(\alpha)$ (i) is the conjugate prior of the multinomial token-likelihood, so collapse is a closed-form update $\alpha \leftarrow \alpha + \#(\text{evidence})$; (ii) retains *tail mass* — no concept is ever assigned probability zero, only low confidence, which is the Bayesian answer to the black-swan objection (a frequency-zero hazard still carries posterior mass; the “never observed, therefore impossible” fallacy); (iii) its concentration parameter α_k is directly interpretable as the activation intensity of concept k , and can be conditioned on context variables (time, place) as latent modulators.

The two axes are complementary, not competing: **bases are the skeleton (A2–A3), the Dirichlet is the weather (this section)**. A probe collapses the distribution (probe $\rightarrow$ collapse $\rightarrow$ KL-verify is the same closed loop our garden runs on exp10/exp13/exp17); the KL divergence between activated and expected concept spaces *is* the falsifiable residue that A3’s combination term predicts. Different agents sharing bases with different coefficients generalizes the RGB metaphor of the companion white paper; the residual off-plane component (44°) is what RGB-with-three-channels would miss and our fourth axis measures.

5 Evidence from the garden

5.1 Composition without concatenation (A3)

Eleven-point mixture scans of LoRA adapters, two independent corpus pairs (spring/festive and triad noir), park hyperparameters, seed 13: angle-sum excess $85^\circ \rightarrow 123^\circ$ at midweight; SSS solution of the endpoint triangle places the midpoint **44.0°** (spring) and **46.5°** (noir) off the geodesic — conserved to 2.5° across domains (C27, C34). Edge swing: 10% of the minority corpus rotates the adapter by $\sim 33^\circ$. Loss and geometry exclude a jump: the mixture path is smooth but *out-of-plane*.

5.2 Ordinal density law (A1, A2)

Three-ball ordinal battery (Ballon d’Or ranks, NBA draft picks, MLB WAR ranking), 3 arms $\times$ 3 text orders: diagonal dominance 9/9, direction index flipping sign strictly with training order, shuffled-arm index ≈ 0 (C29); narrative-order rotation 7° , permutation-order 56° , cross-domain 83° (C28). Reading: *the angle an order channel subtends equals its token density times a family constant* — order is annotation when sparse and half-skeleton when dense.

5.3 Innate spaces, measured by density vs. scale (C31, C35)

Fresh-composition probes (words known, pairings never before spoken; stateless Kaggle kernels as the strictest “new session”): target-space mean rank improves monotonically 0.5B $\rightarrow$ 1.5B $\rightarrow$ 3B (33.4/25.9/15.5 of 172; chance 86), top-3 hit rate 5–10 $\times$ chance. Concept space exists innately and weakly at 0.5B; connectivity grows with scale. The anchor sentence “Coffee is warming, sun is rising” lights up the Red-Pill space at 0.5B (n=1, anecdote logged, not evidence). A subsequent fp16 rerun at 7B *broke* the ladder: mean rank 31.0, worse than 3B’s 15.5. We log the breach instead of explaining it away. Three candidate causes stand — (i) leakage (the 7B pretraining corpus may genuinely contain our “fresh” pairings), (ii) mean-pool drift in wider representation spaces, (iii) genuine non-monotonicity — to be adjudicated by E17b (anti-leakage construction + last-token control) before any rescue of P-B3.

5.4 Centering as re-manifolding (A4)

172-space atlas: raw cosine separates intra/inter by 0.051 (H-A1 fail, anisotropy floor confirmed); after global centering, inter-space mean collapses to 0.0056 and separation reaches 0.677 — a 13-fold gain; strongest coherent spaces are seasonal and noir-film scenes (C35 and atlas-v2 dual-track report). Cross-space angles concentrate near orthogonality (83–90°): manifold patches are quasi-orthogonal, which is the geometric content of “cross-domain is hard”.

5.5 Metaphor as long geodesic (C30)

If “A is B” activates bidirectionally, A and B inhabit one space at large distance: metaphor difficulty = path length, cross-space mapping is a myth of appearance. Audit experiment (E20: co-activation of 15 canonical pairs over the atlas) is queued.

6 Predictions (each falsifiable, each with a number)

P-B1 (residual bound): for any two specialists, mixture arms must retain >25% off-plane fraction; failure falsifies A3’s constitutive residual. P-B2 (orthogonal patches): centered inter-space angles concentrate in $[75^\circ, 100^\circ]$; significant mass below would indicate unmapped shared spaces. P-B3 (density-scale curve): mean fresh-probe rank improves monotonically with parameters on a log scale. *Status: breached at 7B (31.0 vs 3B’s 15.5); under E17b audit — stands falsified unless leakage or pooling artifacts are demonstrated.* P-B4 (metaphor distance): pairs rated “hard” by humans show longer atlas shortest paths than “easy” pairs, measured by probe-cascade hops.

7 Related positions

Tolman’s cognitive map and the grid-cell lineage (neural instantiation of A4); Piaget’s sensorimotor schemes (A2’s developmental cousin); Minsky’s “ability to solve hard problems in constrained environments” (an early rank-raising attempt); manifold hypothesis (Bengio et al.) read here not as a learning assumption but as an ontology of agency; Ebbinghaus’s single-curve memory — refuted by the same rank argument we apply to intelligence verdicts (C23–C25).

8 Limitations

Models 0.5B–7B only (2.4T-class claims unverified, excluded); corpora are small and bilingual (zh-dominant probes); rank spectra depend on the LoRA parameterization ($r=16$ budget); “agent” here carries no phenomenal commitment — this is a geometry of behavior, not a theory of mind; and the ledger’s own law binds us: every number above was frozen before its judgment.

9 Conclusion

Four sentences — interact-and-change, the curve-atom, composition-plus-residual, the manifold — replace an unanswerable binary with measurable spectra. The garden’s data already speak them: 44° of newness in every mixture, 7° of annotation in every story, 13-fold sight after centering, and a rank ladder climbing with scale. Intelligence, demoted from definition and promoted to dimension.

Reproducibility

All experiments: stateless Kaggle kernels (sha-pinned models), PREREG-gated splits, git-ledger timestamps (LETHE/cora-atlas/chora). The kernel *is* the Dolores control: no session, no memory, no mercy.

Appendix — the ledger of every number cited

Ledger	Quantity	Value
C27/C34	mixture bulge; off-geodesic detour	angle-sum $85^\circ \rightarrow 123^\circ$; 44.0° spring / 46.5° noir (SSS, conserved $\pm 2.5^\circ$)
C28	narrative order vs domain	7.0° vs 85.3° ; Δ NLL flip $+0.031/ - 0.016$
C29	permutation order	$56\text{--}60^\circ$; diagonal 9/9; dir-index $+0.42/ - 0.21/ \approx 0$
C35	fresh-probe ladder (of 172; chance 86)	$33.4 \rightarrow 25.9 \rightarrow 15.5$, then 31.0 (breach)
atlas-v2	intra/inter separation	raw 0.051; centered 0.677 (inter 0.0056, 13-fold)

参考文献

- [1] Lola & Yiwei. The Token as a Probe: Geometry of Concept-Space Navigation (v2). cora-atlas/papers/token-as-probe, 2026.
- [2] Letter to Prof. G. Strang (sent 2026-09-20): four subspaces as concept spaces.